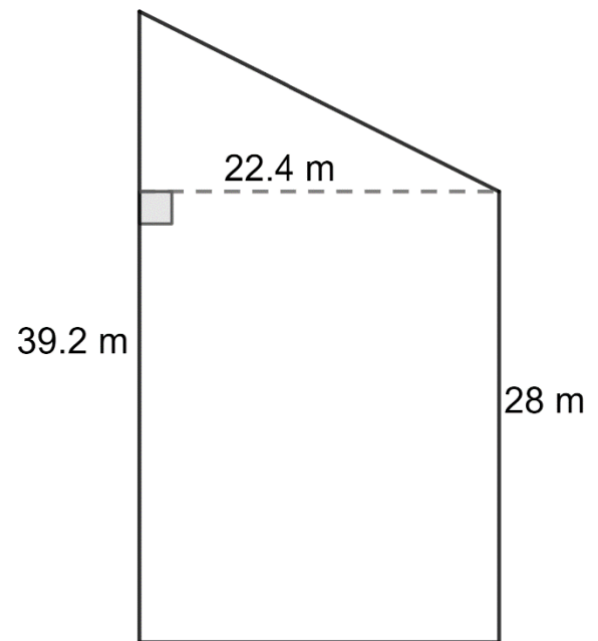


Showing Work with a Calculator: Because you are using a calculator, it is possible to accidentally punch the wrong button. To ensure you earn partial credit, show the operation(s) that you entered into the calculator and the result. It is helpful if you label the operation(s) with information from the word problem.

1. A playground is in the shape of a trapezoid. A diagram of the playground with its dimensions, in meters (m), is shown.

Determine the area of the playground, in square meters (sq m). Show your work.

work	
Area	752.64 sq m



2. The relationship between the

☐ radius
☒ diameter

and the

☐ area
☒ circumference

is proportional because the ratio

be represented by π .

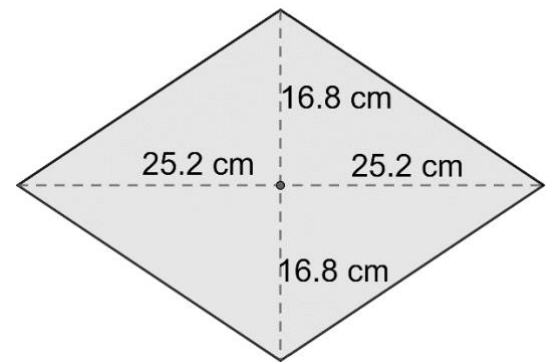
☐ $\frac{A}{d}$
☒ $\frac{C}{d}$
☐ $\frac{A}{r}$
☐ $\frac{C}{r}$

is always the same and can

3. A rhombus is shown with measurements in centimeters (cm).

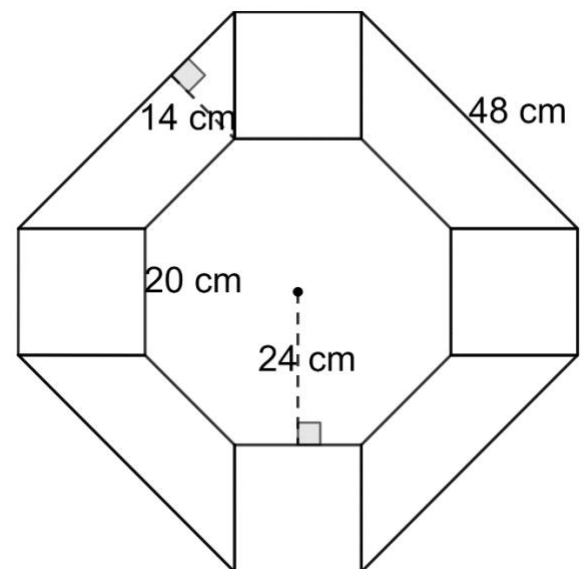
Determine the area, in square centimeters (sq cm), of the rhombus. Show your work.

work	
Area	846.72 sq cm



4. A sidewalk is created using bricks in the shape of a square, regular octagon, and a trapezoid. The bricks are placed in the pattern with measurements in centimeters (cm) as shown. Determine the area, in square centimeters (sq cm), of the pattern shown.

work	
Area	5424 sq cm



5. A teeter totter at a preschool also spins in a circle. The distance from the center to the wheel below one of the seats is 37 inches (in.).
- a. Determine the circumference of the circle that the teeter totter makes as it spins. Round your answer to the nearest tenth of an inch. Show your work.

work	
Circumference	232.6 or 232.5 or 232.4 in.

**Answer depends on which approximation for pi is used.*

- b. The preschool plans to put mulch inside the circle that the teeter totter makes. Determine the area, in square inches (sq in.), where the preschool will place the mulch. Round your answer to the nearest tenth of a square inch. Show your work.

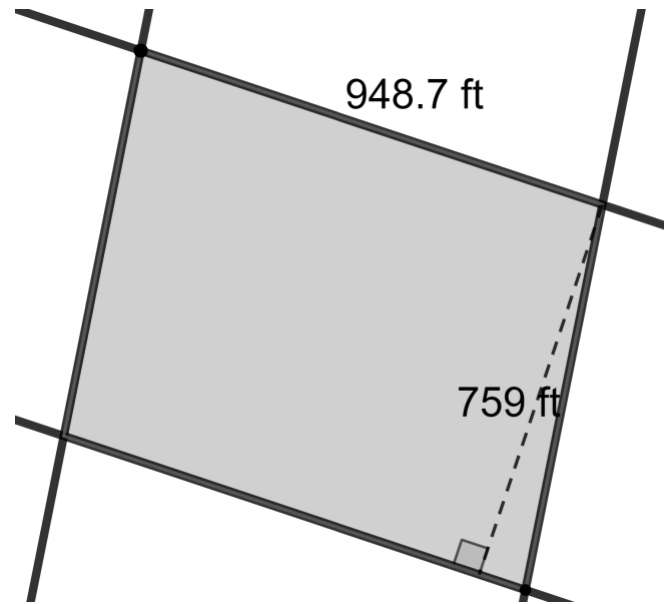
work	
Area	4300.8 or 4302.6 or 4298.7 sq in.

**Answer depends on which approximation for pi is used.*

6. A city planner is redesigning the space between two sets of parallel roads. A diagram of the roads is shown with dimensions given in feet (ft).

Determine the area between the two sets of parallel roads.

work	
Area	720,063.3 sq ft

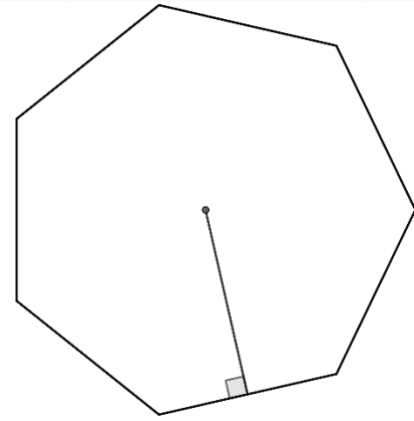


7. Determine the area of a circle, in square millimeters (sq mm) whose radius is 8 millimeters (mm). Leave your answer in pi form.

work	
Area	64π sq mm

8. A heptagon is shown, where each side has a measurement of 102 inches (in.). The perpendicular distance from the center to midpoint of a side is 105.91 in.

Determine the area of the heptagon.



work	
Area	37809.87 sq in

9. Determine the circumference of a circle, in yards, whose radius is $\frac{19}{2}$ yards (yd). Leave your answer in pi form.

work	
Circumference	19π yd